

Classic Sound Synthesis Techniques

Anxe Faraldo





Some Introductory Tensions

“We also have sound-houses,
where we practice and
demonstrate all sounds and
their generation.”

Francis Bacon (1626), *The New Atlantis*

Andrew Hugill (2006, 2017): “The origins of electronic music”. In Nick Collins & Julio d’Escriván (eds): *The Cambridge Companion to Electronic Music*. Cambridge University Press

“The raw material of music is sound.”

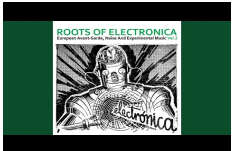
Edgar Varèse (1936), *The Liberation of Sound*

“I need an entirely new medium of expression: a sound-*producing* machine, not a sound-*reproducing* one.”

Edgar Varèse (1939), *Music as an Art-Science*



Arnold Schoenberg (1909) *Fünf Orchesterstücke* #3 Op. 16



Luigi Russolo, *Awakening of a City*

“TO MAKE MUSIC WILL CONTINUE
AND INCREASE UNTIL WE REACH A
MUSIC PRODUCED THROUGH THE
AID OF ELECTRICAL INSTRUMENTS,
WHICH WILL MAKE AVAILABLE FOR
MUSICAL PURPOSES ANY AND ALL
SOUNDS THAT CAN BE HEARD”

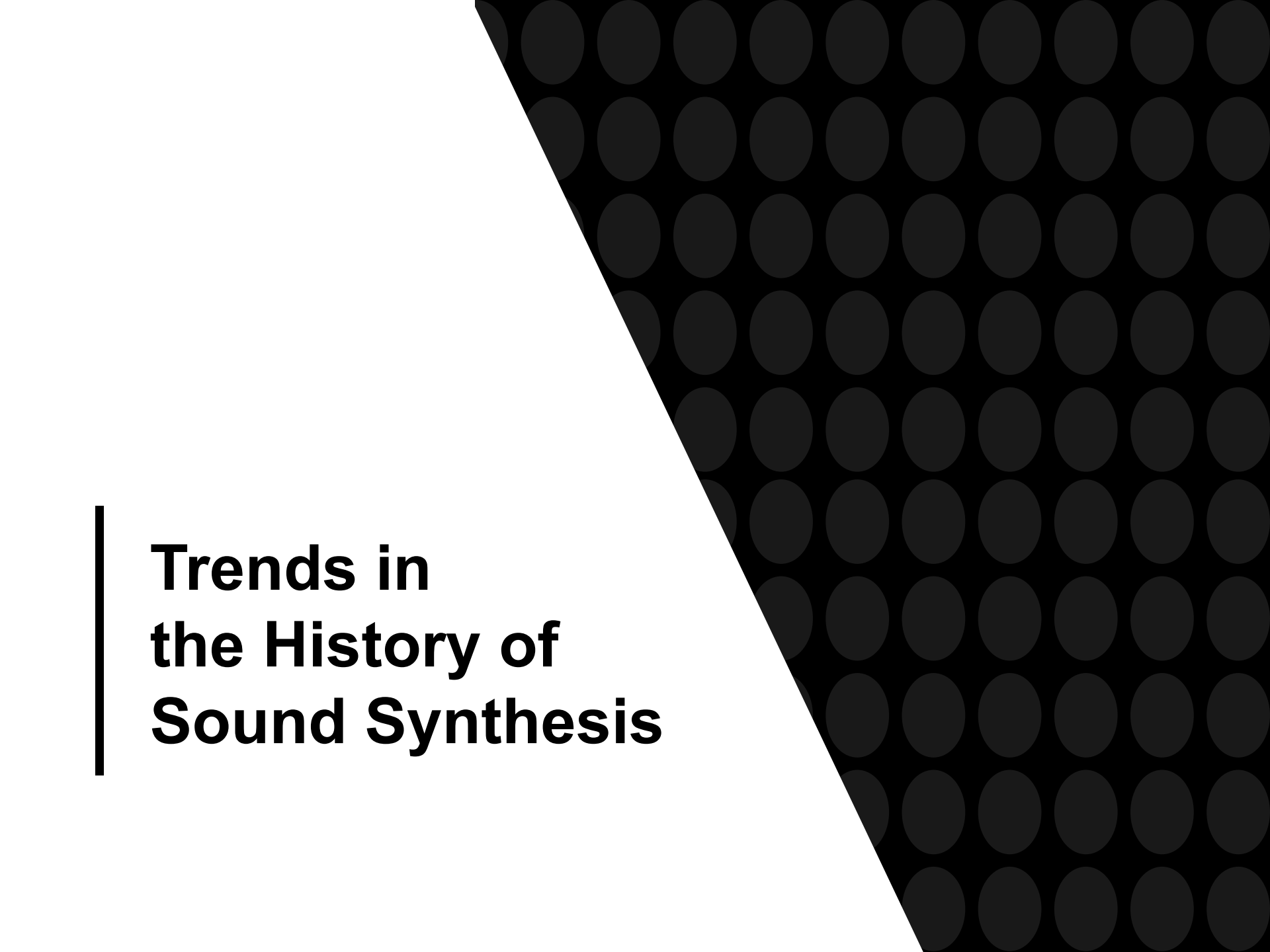
John Cage (1937) *The Future of Music: Credo*

In Christoph Cox and Daniel Warner (2006) *Audio Culture, Readings in Modern Music*. Continuum Books.

“Sound Synthesis is Dead”

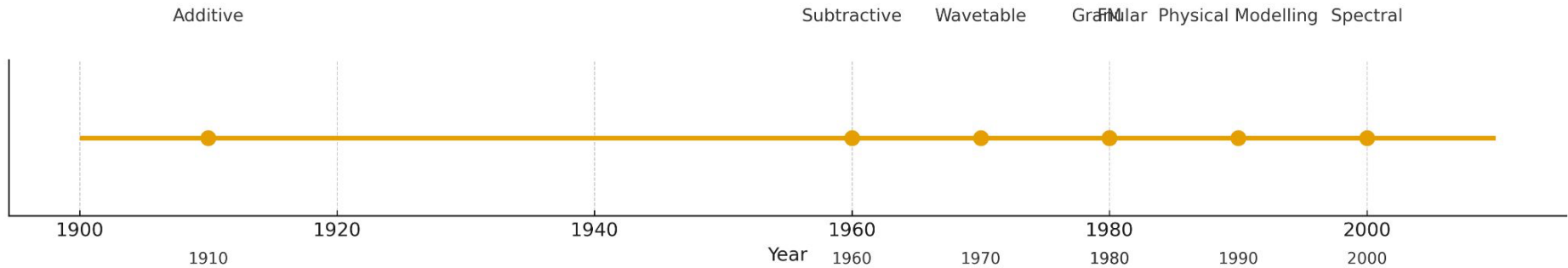
attributed to Xavier Serra (2001) by *Serafin* (2006)

Estefania Serafin (2006, 2017): “Computer Generation and Manipulation of Sound”. In Nick Collins & Julio d’Escriván (eds): *The Cambridge Companion to Electronic Music*. Cambridge University Press

The background of the slide is split diagonally from the top-left to the bottom-right. The upper-left portion is white, and the lower-right portion is black with a repeating pattern of dark gray circles. A thin vertical black line is positioned to the left of the text.

Trends in the History of Sound Synthesis

A Chronology of Sound Synthesis Techniques



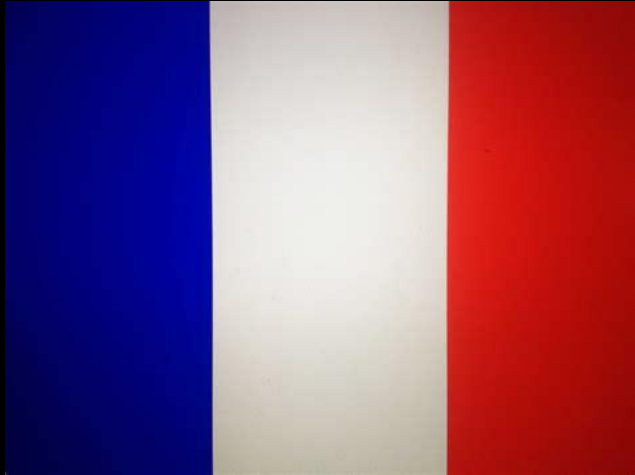


Early Schools of Electronic Sound

Early approaches to electronic music making have pervaded the electronic music history and are still the fundamentals of MOST SOUND SYNTHESIS TECHNIQUES.



The Post War FRENCH Model



a) Sound Recording and Processing Techniques



"The Well Tempered
Microphone"

- c) **Tape Splicing** (cut, paste, reverse, manual envelopes, proto-granular synthesis).
- d) Tape-based **effects** (delays, multi-heads, flangers, choruses...)
- e) Origins of **accumulative compositional forms** (layering)!

The Post War GERMAN Model



Basic electronic components:

- a) **Sound Generators** (oscillators, noise generators)
- b) **Filters**
- c) **Amplifiers**
- d) **Envelope Generators**
 - i) Periodic (Modulators, LFO's)
 - ii) Triggers
- e) **Ring Modulation**

SUBTRACTIVE SYNTHESIS

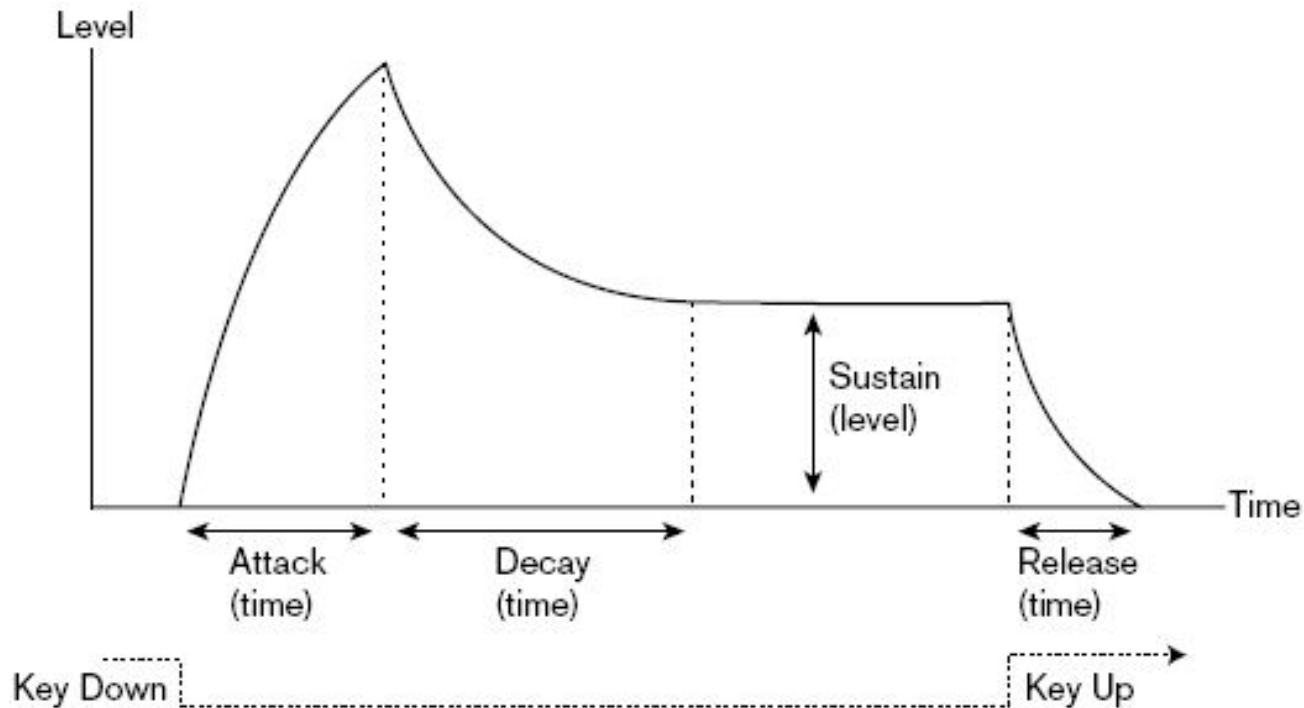
“Classic” Modular Synthesiser Paradigms

Who	Robert Moog	Don Buchla
where	East Coast (NY)	West Coast (CA)
orientation	commercial	experimental
control	piano keyboard	VC sequencer
composer	Walter Carlos	Morton Subotnick
relevant work	 <i>Switched on Bach</i> (1968)	 <i>Silver Apples of the Moon</i> (1967)

Envelopes

Typical growth and extinction curves used for controlling the amplitude of a signal and other sound parameters.

Human ear tends to prefer exponential curves (as opposed to linear).



Evolutions of some trends

**Additive Synthesis >>> Wavetable Synthesis >>>
Analysis and Re-synthesis (Fourier) >>> Concatenative**

**Musique concrète >>> Tape Music Manipulations >>>
Sampling >>> Granular >>> Fixed Media???**

**Elektronische Musik >>> Subtractive Synthesis >>>
Modular Synthesisers >>> Live electronics-coding???**

Genuinely Digital Techniques

- **Fourier** analysis >>> “**Analytical** additive synthesis >>> Jean-Claude Risset >>> **Spectral Model** Synthesis (mimic timbre)
- **FM Synthesis** (Chowning, 1973)
- **Granular Synthesis** (Xenakis, Roads)
- “**Number Crunching**” Techniques >>> Gottfried Michael Koenig, Barry Truax, Iannis Xenakis, Sergio Luque >>> **Stochastic Synthesis.**
- **Physical** Models (mimic sound-production techniques)

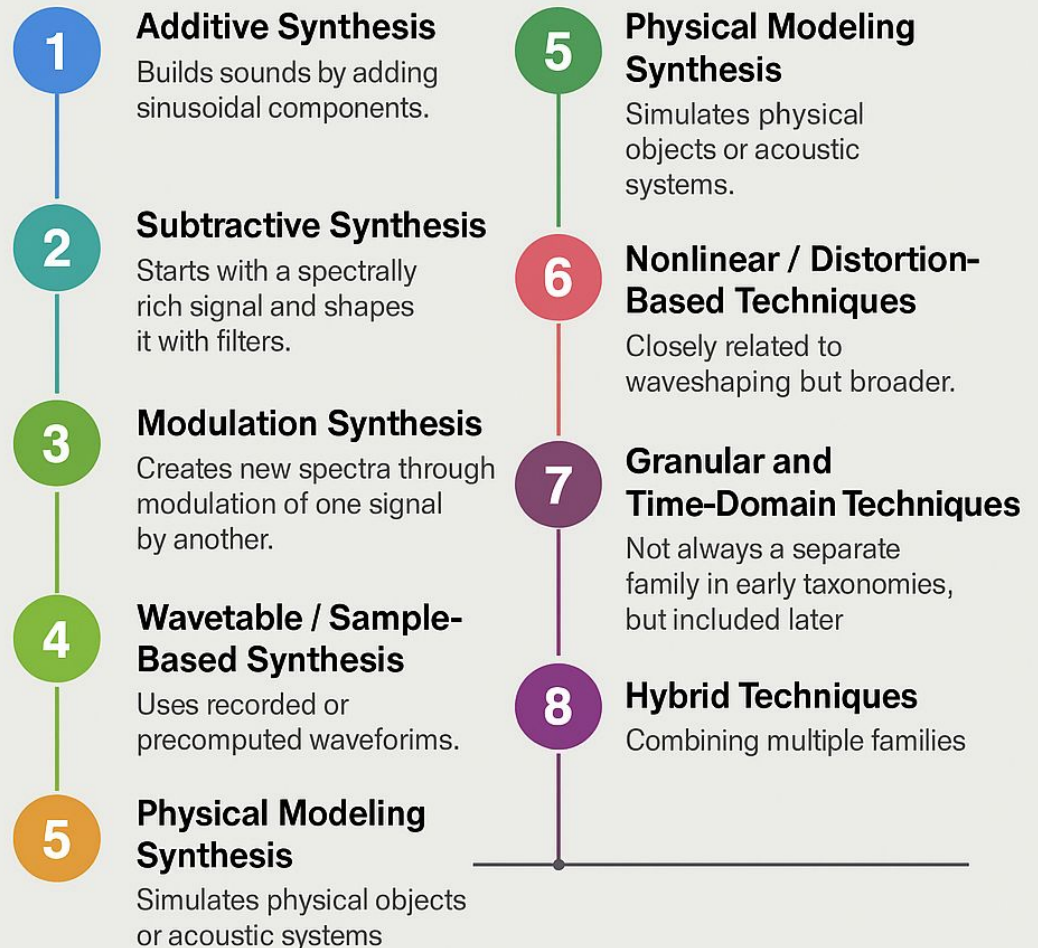
Julius O. Smith III's **Digital** Synthesis Taxonomy

- Processed Recordings
 - Abstract Algorithms (wave-shaping, FM, etc.)
 - Spectral Models
 - Physical Models
- 
- Sound Models

Julius O. Smith III (1991) "Viewpoints on the History of Digital Synthesis". *Proc. of the International Computer Music Conference*.

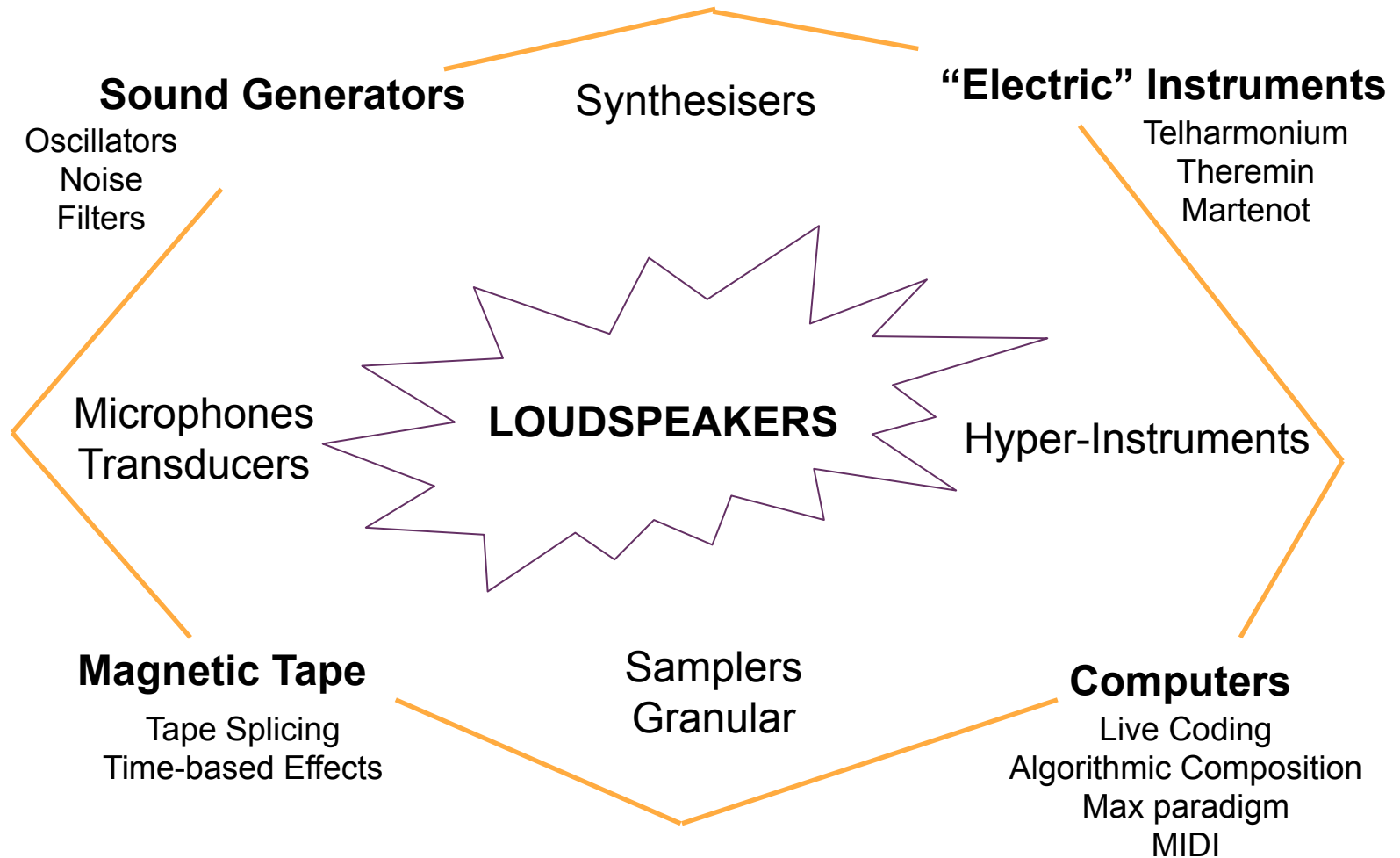
Julius O. Smith III's Digital Synthesis Taxonomy

Julius O. Smith – Digital Synthesis Taxonomy

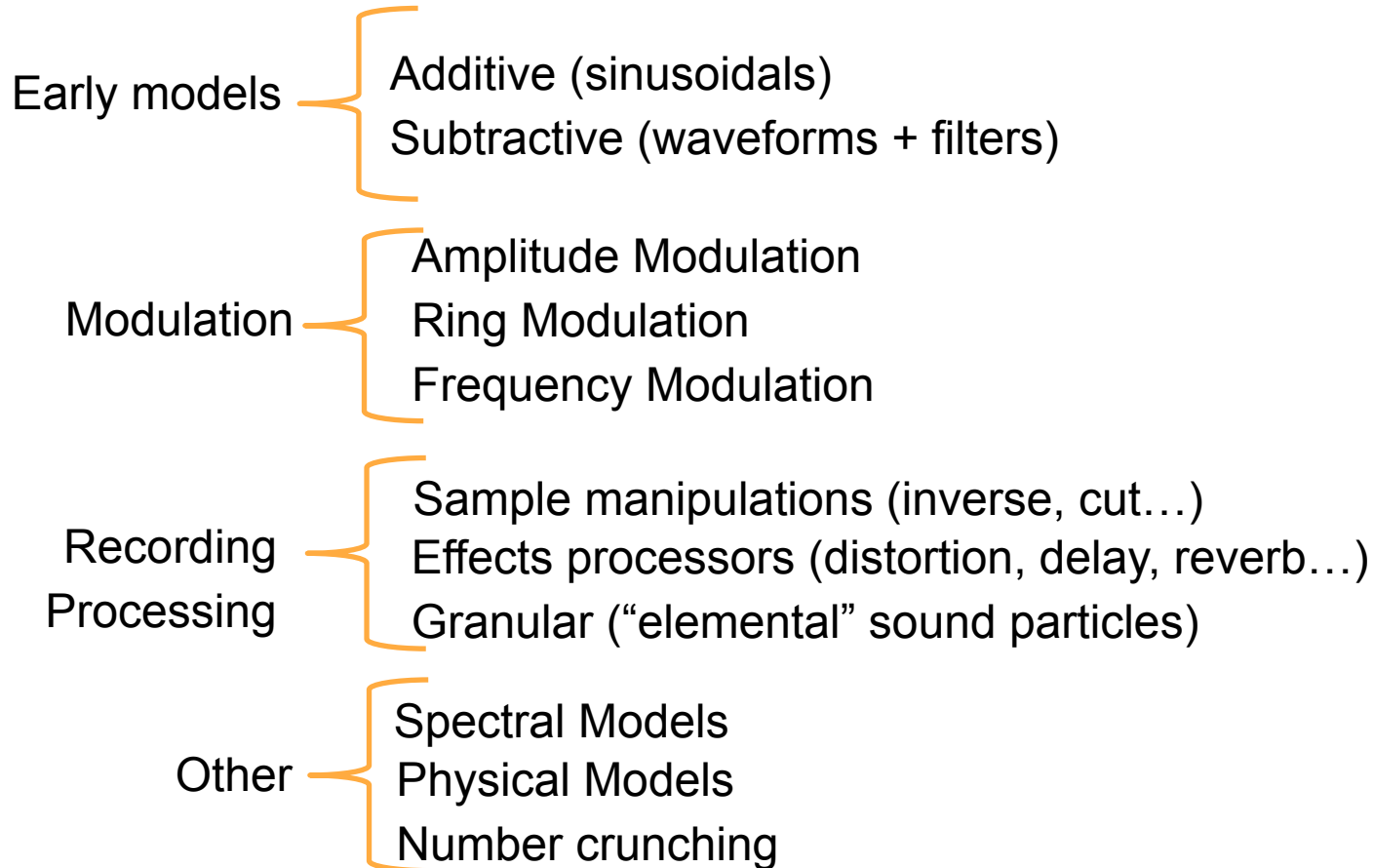


Julius O. Smith III (1991) "Viewpoints on the History of Digital Synthesis". *Proc. of the International Computer Music Conference*.

The “Orchestra” of Electronic Sound



Summary of Synthesis Techniques

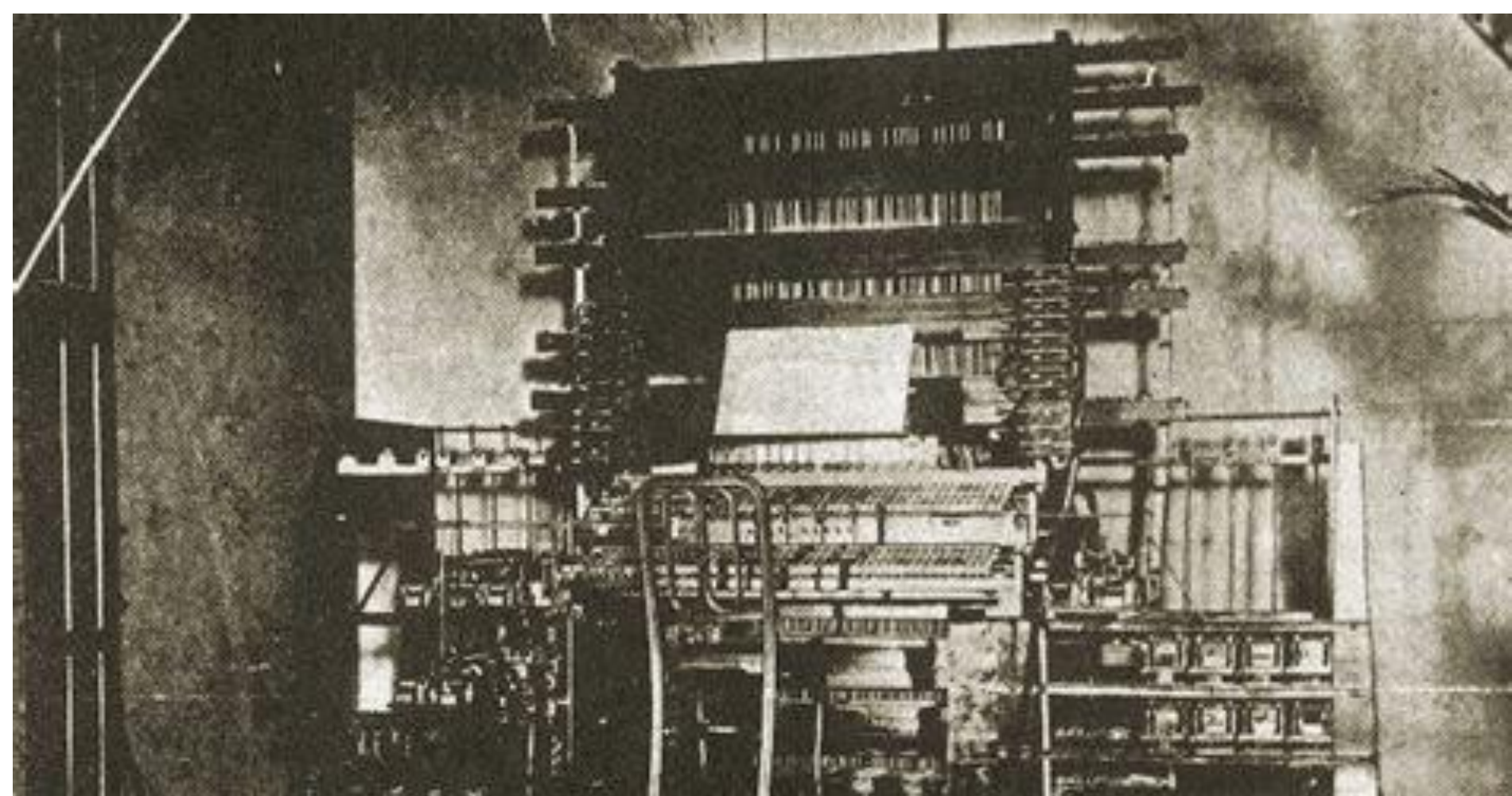




Review of Classic Synthesis Techniques

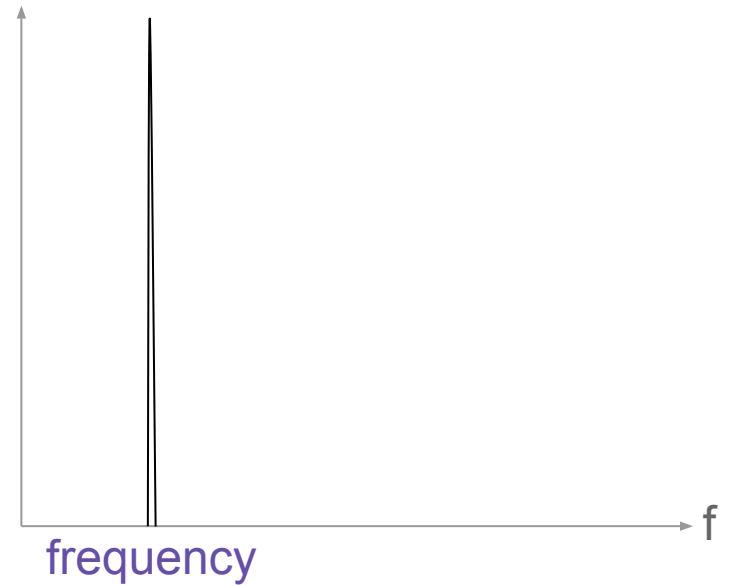
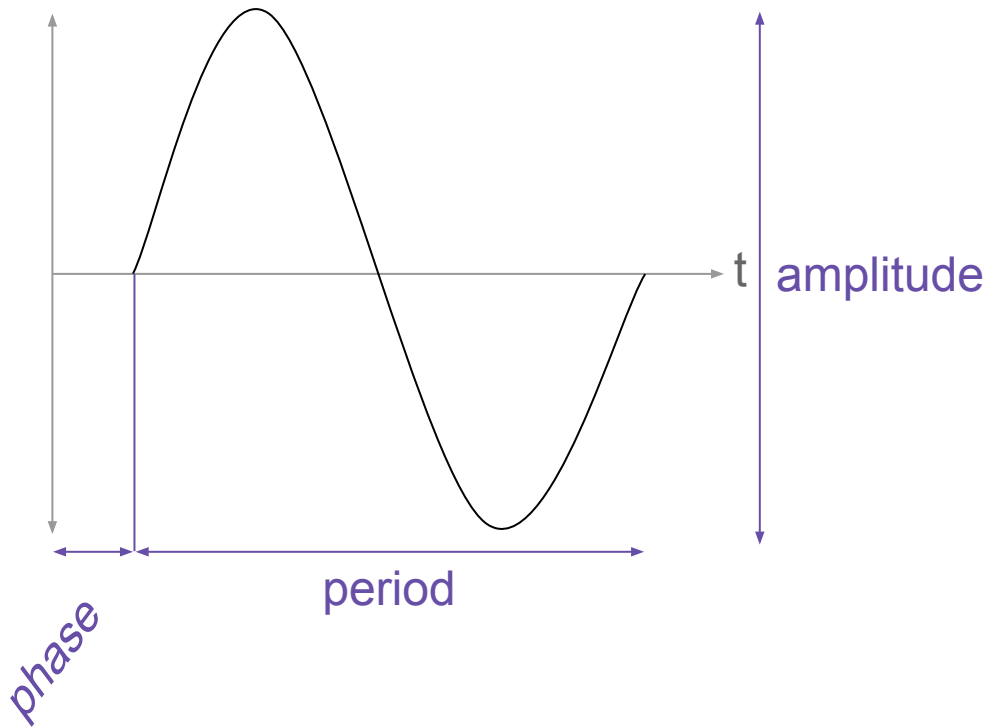
Additive Synthesis







Sine Waves: The Simplest Tones Imagined



Sine aggregation results in complex tones

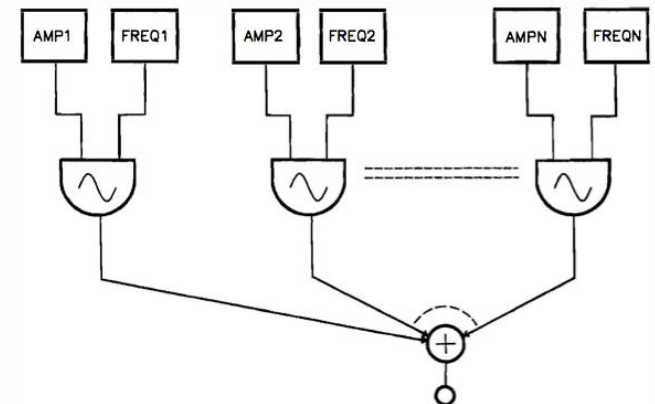
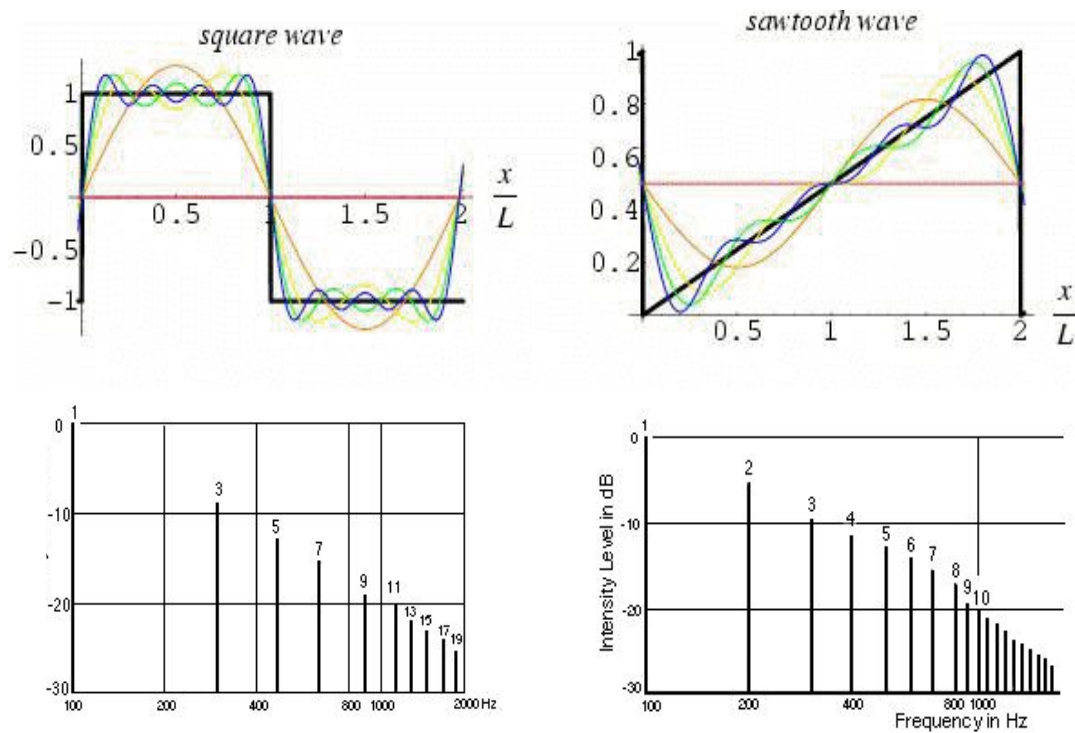
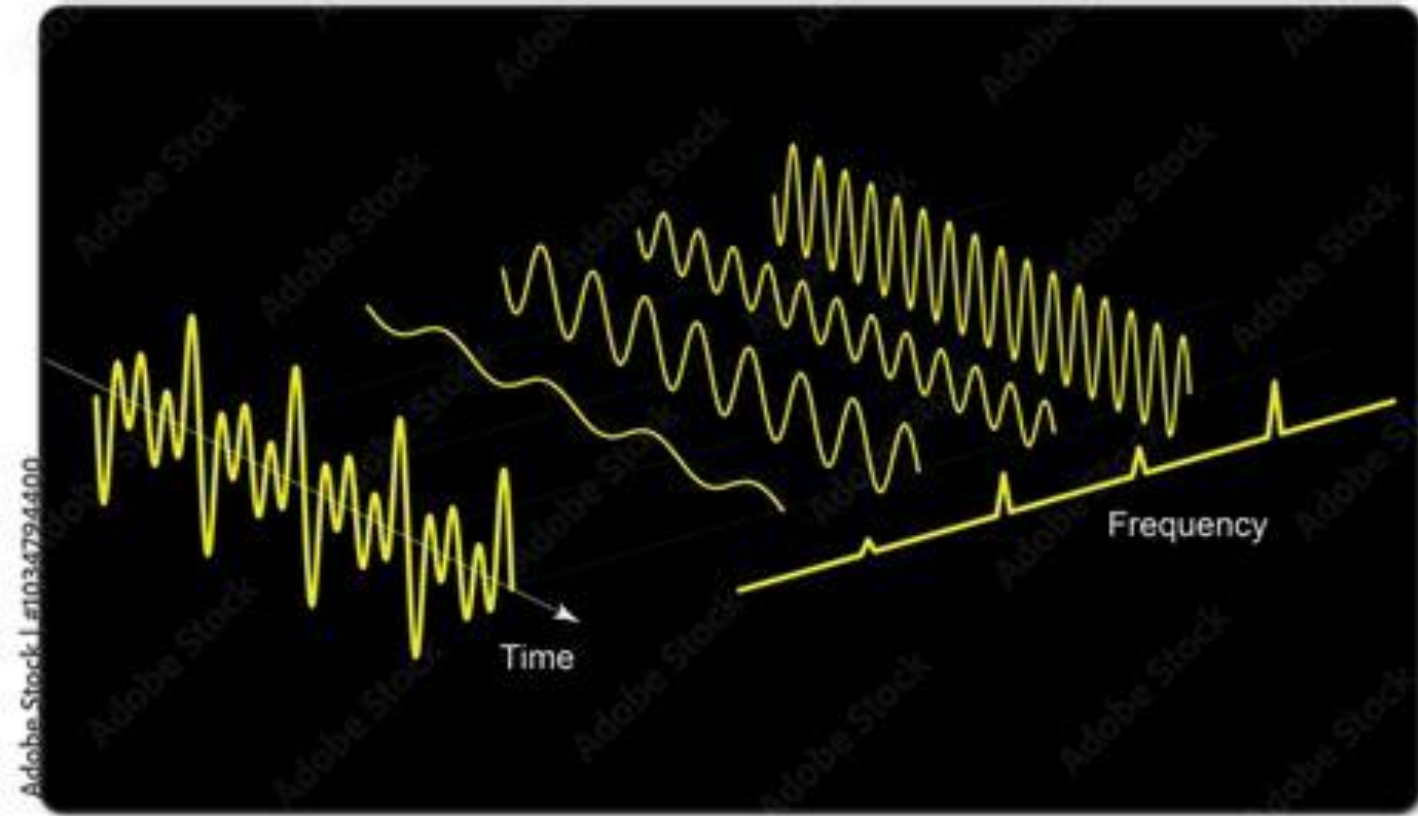


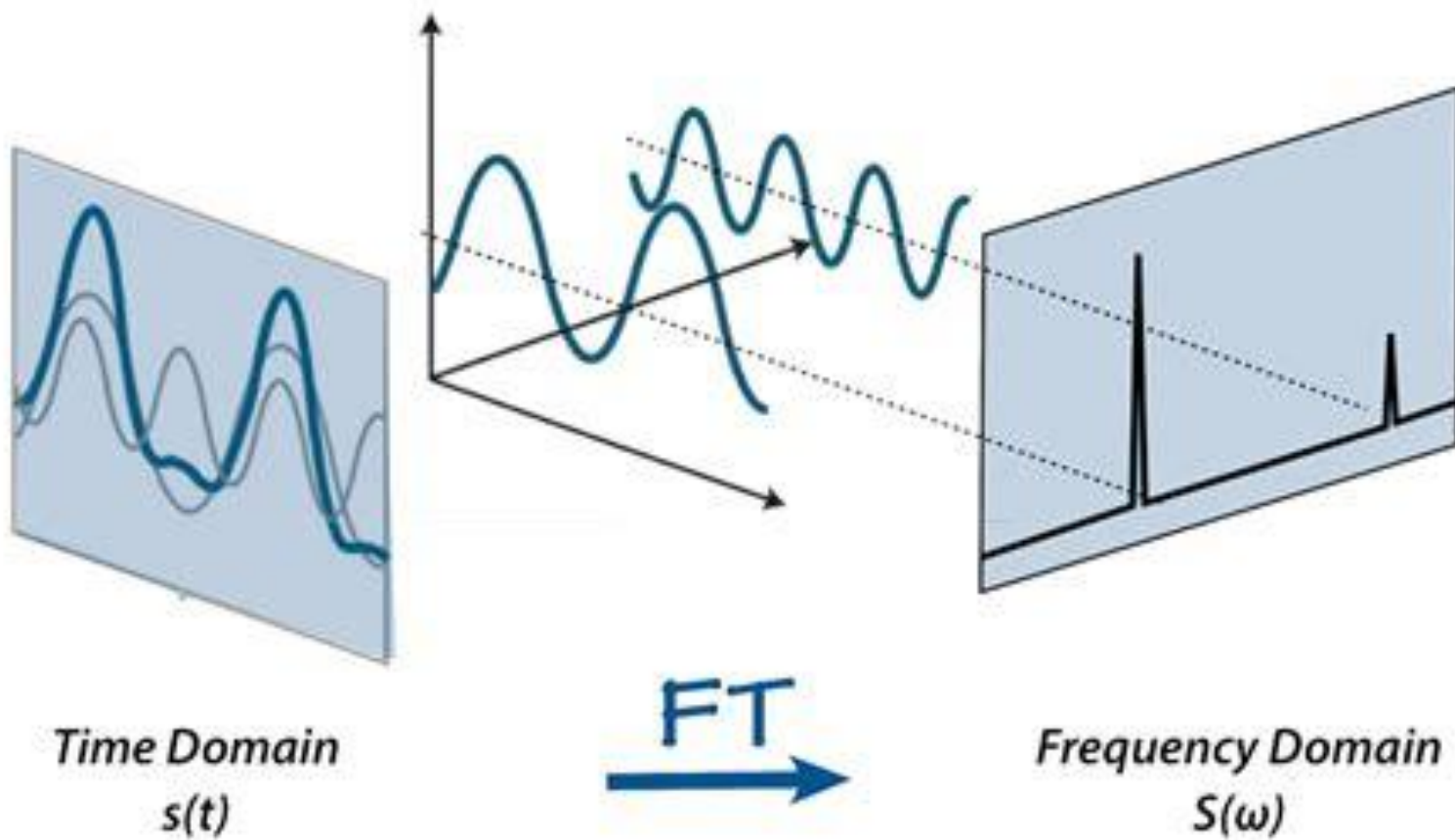
FIGURE 4.12 Basic configuration for additive synthesis. The amplitude and frequency inputs of each oscillator derive from independent function generators, which usually take the form of envelope generators or oscillators.

The Fourier Theorem (published in 1827)

Any periodic signal can be constructed with sinusoids.



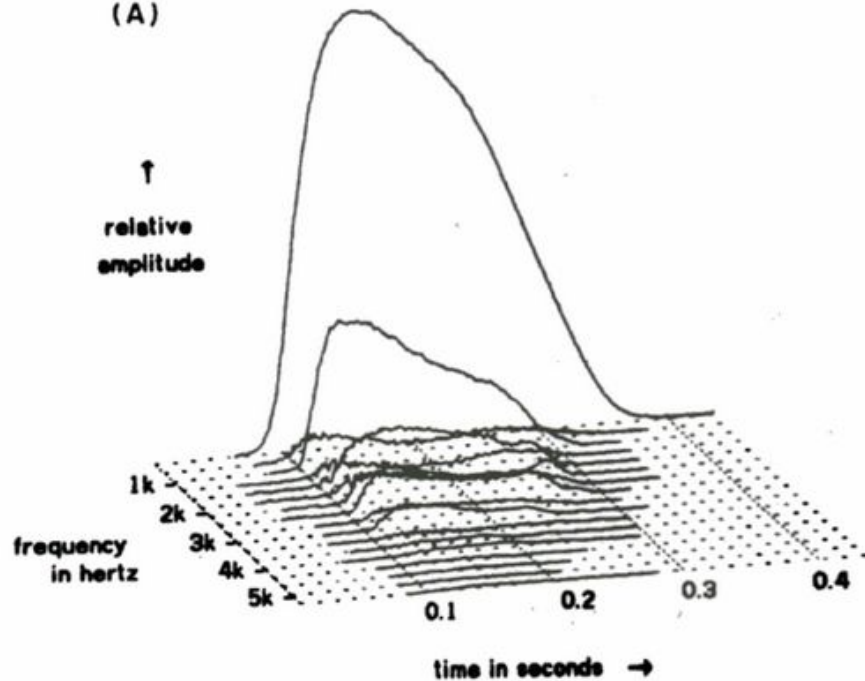
The Fourier Transform



Sinusoidal Decomposition

An oboe

(A)



A trumpet

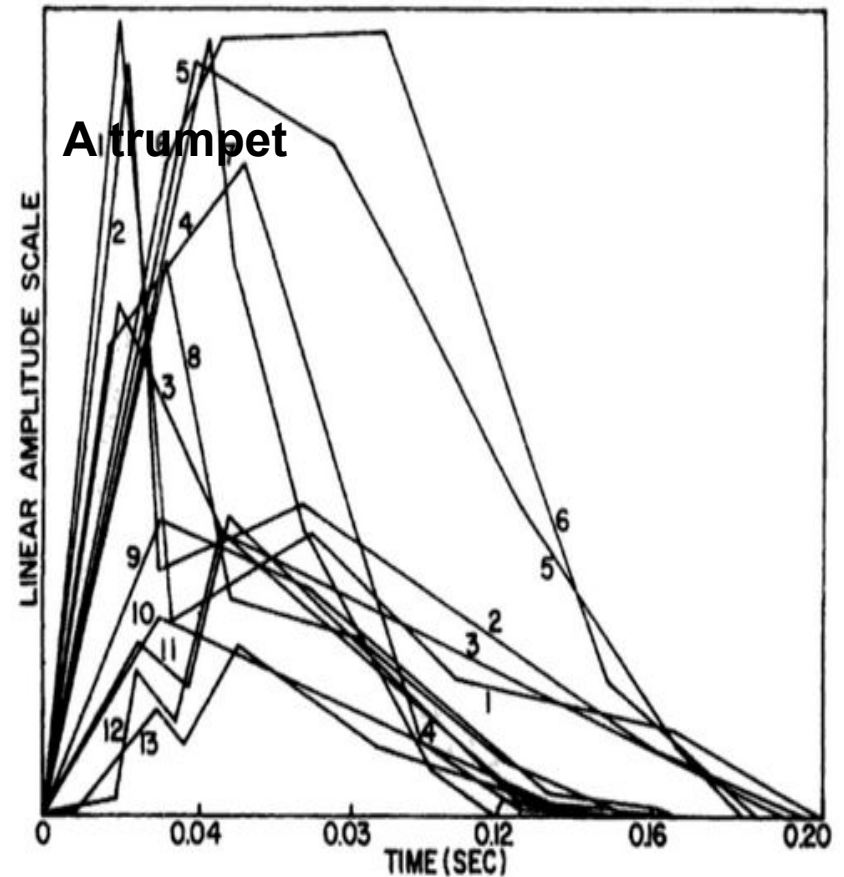
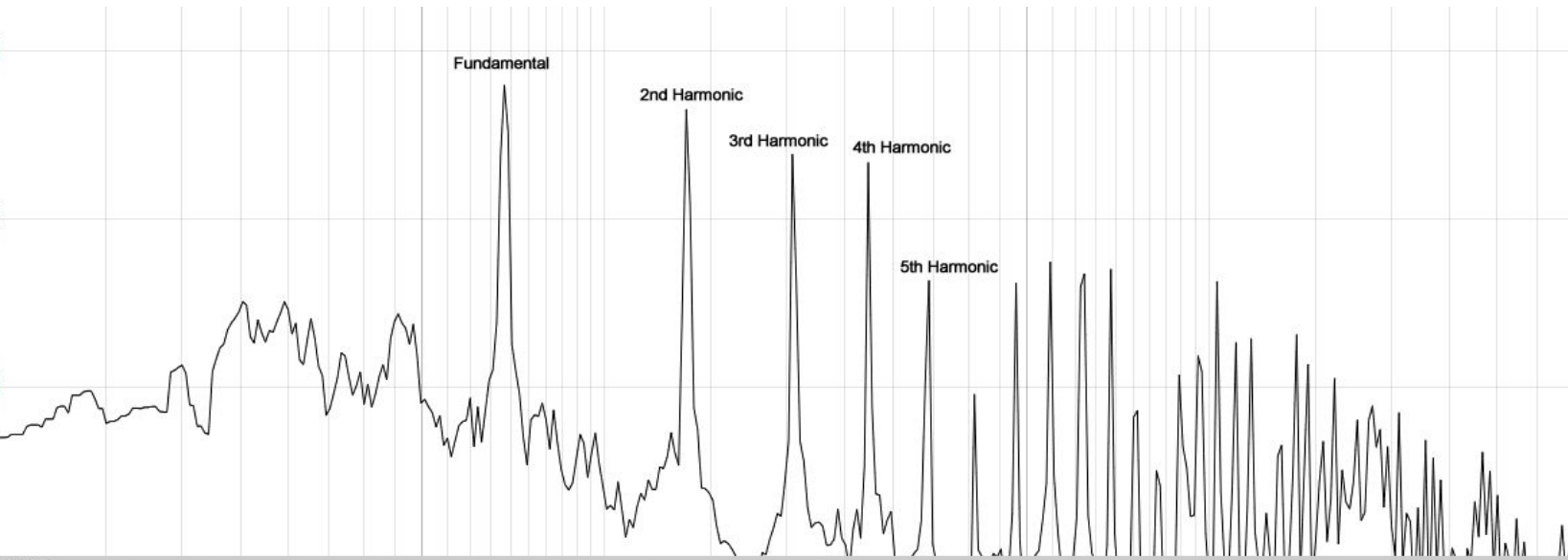
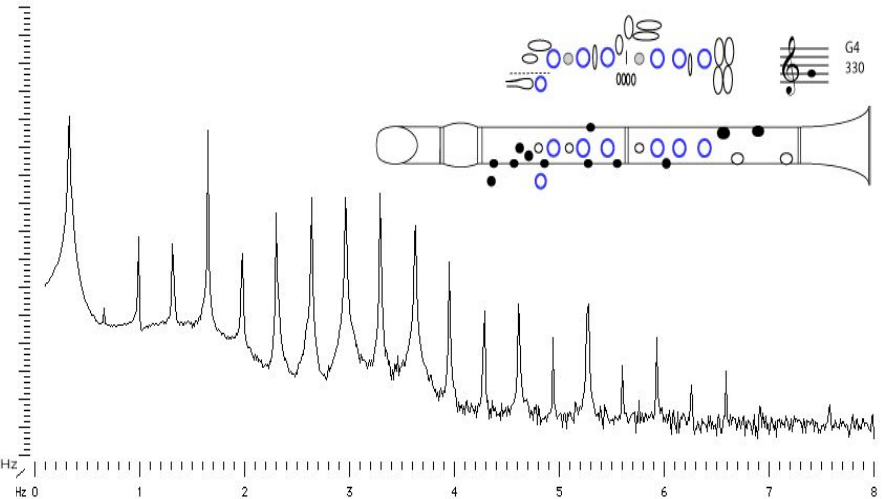
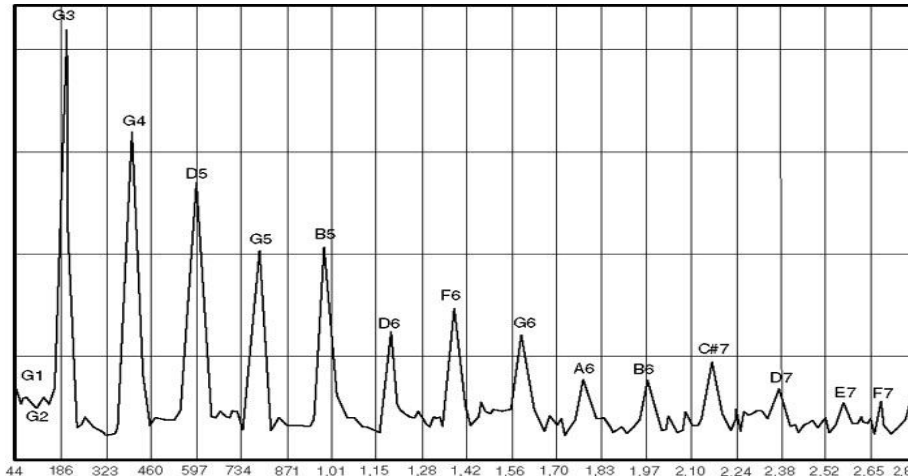
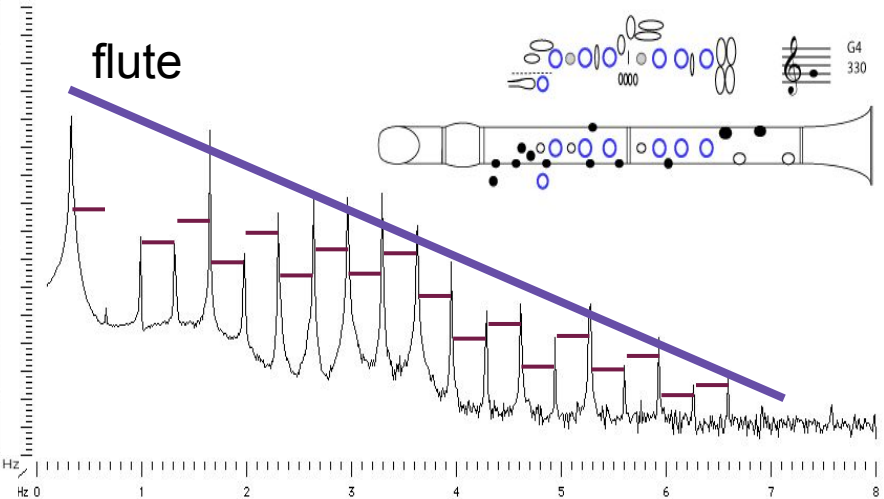
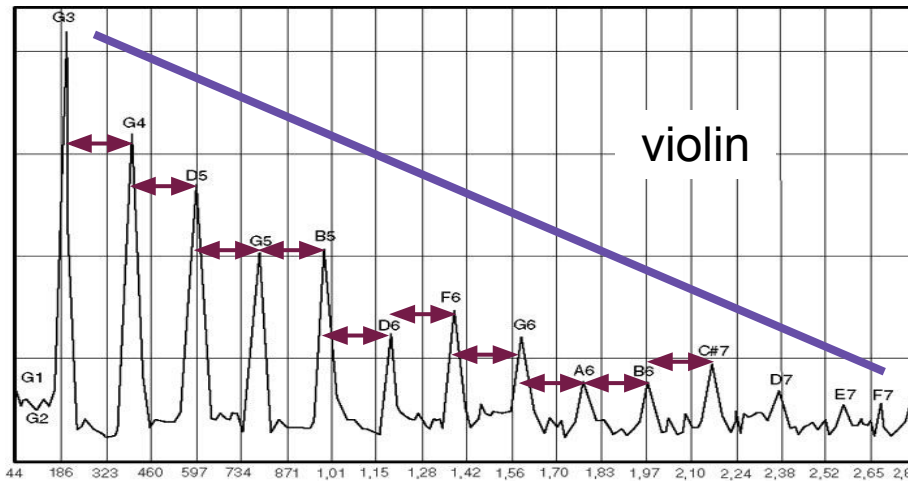


FIGURE 3 Line-segment functions that approximate the evolution in time of 13 harmonics of a D₄ trumpet tone lasting 0.2 sec. Functions like these, obtained by analysis of real tones, have been used to control the harmonic amplitudes of synthetic tones (Risset & Mathews, 1969).

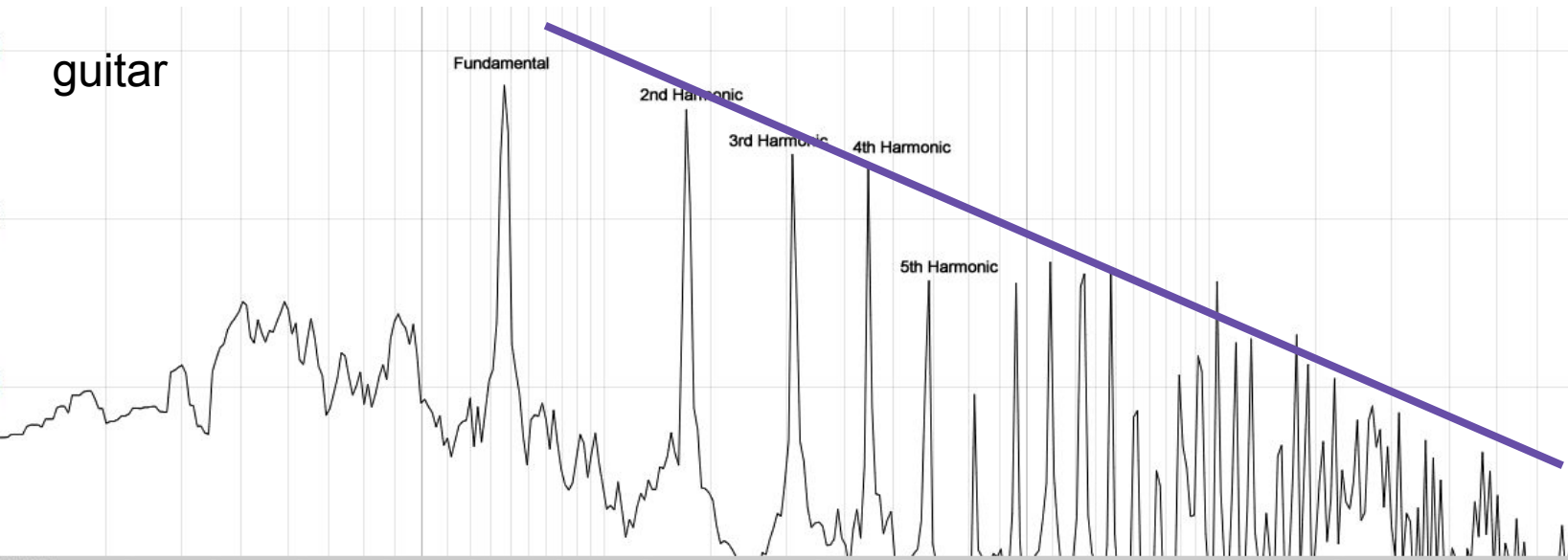
The Characteristic Sound of Instruments: Timbre



Timbre: Harmonics and Spectral Envelope

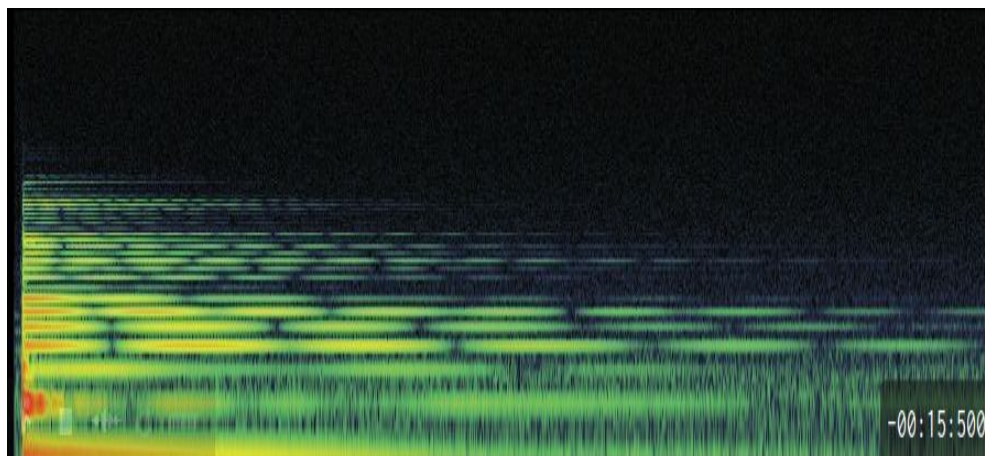


guitar

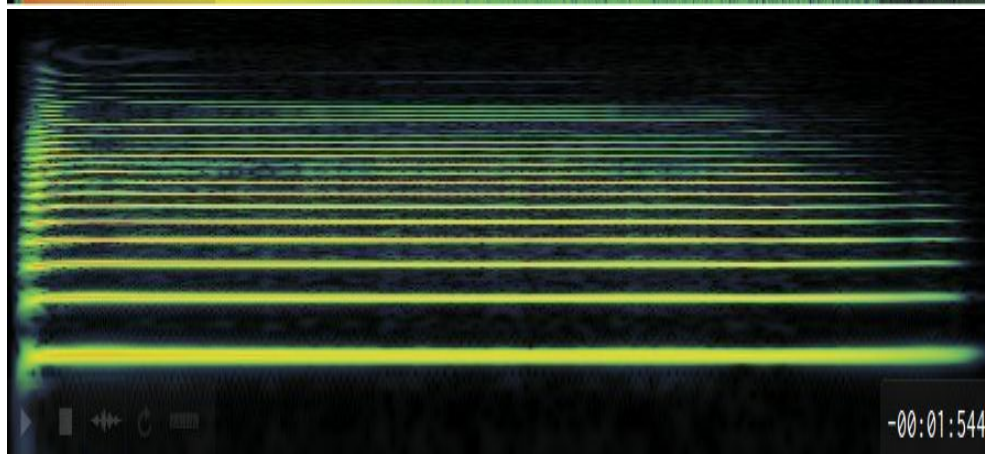


Spectral Envelope Changes in Time

Piano G2



Trumpet G2



Our perception of timbre is deeply related with how spectral components change in the short time: in the millisecond range. Subtle changes in amplitude frequency and phase affect timbre sensation.

Timbre vs. Sound Synthesis

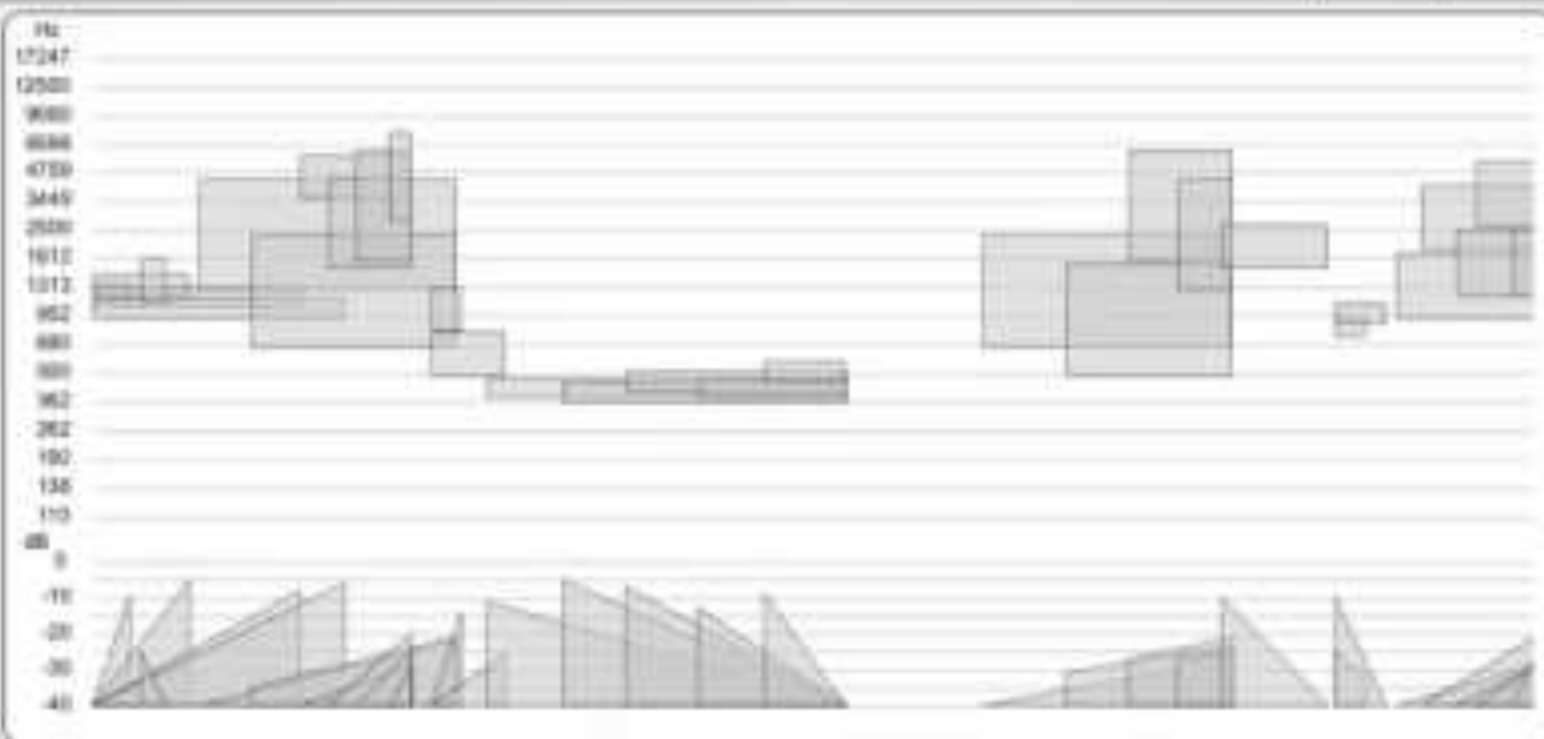


Sound synthesis refers to the generation and manipulation of the components of a sound signal.



Jonathan Harvey - Mortuous Plango Vivos Voco

Karlheinz Stockhausen: Elektronische STUDIE II (1954)



Control